RETAILVISION AI

2.0

THE ADAPTIVE STORE BRAIN

A closed-loop AI system that does not only tell a supermarket what happened - it learns which store action creates incremental profit, tests it safely, verifies the result, and remembers it.

SEE	THINK	TEST	LEARN
Camera + POS + inventory	Causal diagnosis	Micro-experiments	Store-specific memory

IMPORTANT NOVELTY NOTE

This blueprint does not claim that the idea is globally unprecedented. A September 17, 2026 review of publicly documented products found substantial overlap in traffic analytics, product interaction, shelf intelligence, store digital twins, and retail experimentation. The proposed white-space is the combination: retrofit sensing + causal action selection + pre-test simulation + intervention verification + persistent store-specific policy learning in one closed loop. That combination must still be validated through customer discovery, competitive research, and patent review.

Manager-ready version | Prepared for pilot and startup evaluation

01 | EXECUTIVE DECISION MEMO

The stronger company is not a camera company

The upgrade is from analytics -> autonomous learning loop.

The original RetailVision concept is already strong as a practical store-intelligence product: connect existing cameras with POS and inventory, identify stockouts and shopper behavior, and turn those signals into manager actions. The market review shows, however, that several parts of that feature set are already publicly offered by established companies such as Trigo, RetailNext, Vusion, Shopic and others. [W1-W5]

The strategic upgrade is therefore to define RetailVision around a harder problem: which operational action actually causes incremental value at this specific store, under its own constraints, and can the system learn that answer over time?

NEW CORE THESIS

RetailVision becomes an AI Store Scientist: it observes the store, forms a hypothesis, estimates whether an intervention is worth trying, runs a low-risk experiment, measures the incremental effect, and updates the store policy. The product compounds learning instead of only producing reports.

1	5	4	1
Closed-loop workflow	Core data layers	Evidence gates	Store memory

What changes from the original blueprint

Original emphasis	2.0 emphasis	Why this matters
Detect events	Detect + decide + test	Computer vision is increasingly commoditized; decision quality is the moat.
Recommend actions	Choose the best next experiment	Avoids generic “you should optimize this” advice.
Dashboard	Action queue + experiment ledger	Managers see what to do, expected value, and whether it worked.
One-time ROI	Cumulative learning asset	Each intervention improves future decisions for the store.
Broad roadmap	Narrow core + expansion modules	Keeps MVP buildable while creating a defensible long-term thesis.

02 | COMPETITIVE REALITY

What already exists - and where the gap may be

This is the part that makes the idea credible rather than exaggerated.

The existing blueprint described traffic, dwell, pick/return, shelf gaps, POS correlation, queue analytics and action recommendations. Current public product pages confirm that many of these are already commercial categories. Trigo documents shopper journeys, product interactions, pick/put-back, decision sequence and placement impact. RetailNext documents traffic, conversion, journey analytics and test-and-learn. Vusion documents computer vision for shelf gaps, planograms, inventory and prioritized store tasks. Shopic documents shopper behavior, shelf intelligence, promotions and store analytics. [W1-W4]

Company/category	Publicly documented capability	RetailVision lesson
Trigo Retail	Shopper journey, product interaction, pick/put-back, placement impact, anonymous tracking. [W1]	Do not claim basic behavioral CV as unique.
RetailNext	Traffic, conversion, full shopper path, AI forecasting, prescriptive insights, test-and-learn. [W2]	“Dashboard + recommendations” is not enough.
Vusion	Shelf CV, out-of-stocks, planograms, inventory, prioritized tasks, connected-store intelligence. [W3]	Shelf execution + action is established.
Shopic	Item recognition, shopper decisions, promotions, store analytics, live inventory. [W4]	Shopper + shelf + promotion intelligence already exists.
Digital-twin research / Lowe’s	Physical-store digital twins and simulated placement/layout analysis have been demonstrated publicly. [W6-W7]	A digital twin alone is not the moat.
Grocery operating systems	Vori describes an end-to-end grocery OS spanning payments, shelf tracking, ordering, pricing and promotions. [W8]	Broad “self-driving grocery” positioning is crowded.

WHITE-SPACE HYPOTHESIS

I did not find a public product page in this review that clearly documents the exact full combination of: existing-CCTV retrofit + POS/inventory fusion + automated experiment selection + counterfactual pre-screening + post-intervention causal verification + persistent store-specific action policy. This is a research finding, not proof of global uniqueness. Competitors may have similar capabilities that are private, bundled, or not publicly described.

03 | THE DIFFERENTIATED IDEA

RetailVision AI 2.0: The Adaptive Store Brain

Build the system around a learning loop, not a feature list.

The six-step closed loop

Step	System behavior	Example
1. Observe	Fuse anonymous camera events with POS, inventory, price, promo and staffing context.	“12 shoppers picked SKU A; 8 returned it; shelf was full; price changed yesterday.”
2. Diagnose	Generate competing hypotheses rather than a single explanation.	“Price friction, poor placement, competitor substitution, or packaging confusion.”
3. Pre-screen	Use a store model / synthetic shoppers to reject obviously weak interventions.	“Moving A one bay closer to B is predicted to increase paired discovery.”
4. Experiment	Choose a low-risk test with a control condition and guardrails.	“Test placement in one zone for 7 trading days.”
5. Verify	Measure incremental outcome against baseline/control and log confounders.	“+6.2% units vs matched control; promo excluded.”
6. Learn	Update the store-specific policy memory so the next recommendation improves.	“This store responds better to adjacency than to endcap exposure for this category.”

PRODUCT PROMISE

Every recommendation should carry an expected value, confidence, intervention cost, risk flag, measurement plan, and a post-action result. That makes the system accountable.

The new product name inside the platform

Store Scientist Engine (SSE): the module that turns store data into hypotheses, experiments and verified decisions.

- Observation layer: camera events, shelf state, POS, inventory, price, promotion, staffing and calendar context.
- Diagnosis layer: root-cause hypotheses, substitution patterns, demand anomalies and constraint checks.
- Experiment planner: selects test/control zones, duration, KPI, guardrails and expected value.
- Counterfactual sandbox: pre-screens layouts and interventions using a digital representation of the store and synthetic shoppers.
- Outcome ledger: stores what changed, what the model expected, what happened, and what the evidence says.
- Policy memory: learns which actions work for this store, category, daypart and operating condition.

04 | THE MOAT

Five assets that get stronger with every real-world intervention

The goal is to create a proprietary learning system, not a better object detector.

Asset	What accumulates	Why it becomes hard to copy
Store Event Graph	Time-stamped links among shopper flow, shelf state, SKU, zone, transaction and intervention.	Store-specific history is unique and becomes richer with deployment.
Action-Outcome Ledger	Every recommendation paired with expected lift, actual lift, cost, risk and confounders.	Creates a rare dataset of decisions and measured consequences.
Policy Memory	Rules/models for “what works here” by zone, category, daypart and constraint.	A rival can copy features but not historical cause/effect relationships.
Deployment Graph	Camera geometry, shelf map, SKU map, connector health and calibration history.	Reduces future deployment time and support burden.
Benchmark Network	Aggregated, permissioned benchmarks across store formats.	Creates scale advantages without requiring personally identifying data.

DEFENSIBILITY TEST

Do not call any of these a moat until they are populated by differentiated customer data and repeatedly used in production. The original blueprint correctly treated the moat as something to earn, not claim. [Source blueprint]

Potential patentable areas to investigate

- Automated selection of in-store intervention tests from fused camera/POS/inventory signals under operational constraints.
- Linking physical-store event graphs to an intervention ledger that updates a store-specific decision policy.
- Pre-screening physical-retail interventions in a digital/synthetic shopper environment and converting accepted simulations into controlled real-world tests.
- Confidence-aware action ranking that chooses between “act now”, “run a test”, and “do not act” states.

LEGAL BOUNDARY

These are invention themes for a patent search, not legal conclusions about novelty or patentability. A patent professional should search prior art before any filing.

05 | PRODUCT ARCHITECTURE

From store sensors to a self-improving decision loop

Edge-first, privacy-aware and modular.

Layer	Function	Candidate technology
Physical sensing	Existing CCTV/RTSP, shelf imagery, optional IoT.	RTSP, OpenCV, edge GPU/NPU
Perception	People counting, anonymous tracking, shelf state, pick/return events.	YOLO-family detector, tracker, PyTorch/TensorRT
Commerce context	POS transactions, SKU, inventory, price, promotions.	APIs/files, FastAPI, PostgreSQL
Store model	Zones, shelf geometry, product graph, operational constraints.	PostgreSQL + PostGIS; graph layer optional
Causal engine	Control selection, incrementality estimation, confounder handling.	Python; Bayesian / causal inference methods
Experiment planner	Intervention design, guardrails, task creation.	Rules + optimization + LLM orchestration
Learning memory	Store policy updates, outcome history, model registry.	Feature store / event store / model registry
Experience	Manager queue, experiment status, daily brief, evidence report.	React / Next.js
Cloud control plane	Multi-store configuration, benchmarks, monitoring.	AWS / GCP / Azure

Core data object: the Intervention Record

Field	Example
Hypothesis	“Shelf position is suppressing discovery of SKU A during 5-8 PM.”
Evidence	Low dwell + high pick-return + strong sales of adjacent SKU B.
Intervention	Move SKU A one bay; keep price and promo constant.
Expected effect	Higher discovery and basket attachment.
Guardrails	No blocked emergency path; no change during promo event.
Test design	Test zone vs matched control zone for 7 days.
Observed result	Incremental units, margin, labor minutes, stockout frequency.
Decision	Keep, roll back, or test a stronger variant.
Confidence	High / medium / low with reason.

06 | AI AGENTS

Give each retail decision a job, not a chatbot

LangGraph-style orchestration fits naturally here.

Agent	Responsibility	Output
Watcher Agent	Monitors event streams and detects anomalies.	Candidate situations requiring attention.
Diagnostician Agent	Generates competing hypotheses and checks evidence.	Ranked root-cause hypotheses.
Experiment Agent	Designs control/test setup and guardrails.	Experiment plan.
Economics Agent	Estimates expected incremental revenue/margin and intervention cost.	Expected value band.
Operations Agent	Checks whether the store can actually execute the change.	Feasibility + staffing/task plan.
Verifier Agent	Measures post-action effect and confounders.	Causal evidence summary.
Memory Agent	Updates store-specific policy memory.	New rule/learning entry.
Manager Agent	Explains decision in natural language.	3-5 prioritized actions or tests.

IMPORTANT DESIGN RULE

The LLM should orchestrate and explain; it should not be trusted to invent business facts or calculate the final KPI without deterministic tools. Numeric forecasts, experiment assignment, event counts and financial calculations should come from controlled systems.

Example agent trace

Watcher -> “High pick/return anomaly for cereal SKU A.” Diagnostician -> “Possible price/placement/substitution.” Economics -> “Potential weekly margin opportunity: $X-$Y, low confidence.” Experiment Agent -> “Run placement test first; cheaper and reversible than price intervention.” Operations -> “Store can execute after closing.” Verifier -> “Compare with matched control.” Memory -> “Update adjacency policy if effect persists.”

07 | MVP

Build one loop that can create proof

Do not build a full supermarket operating system in version one.

Must build	Why	Defer
Camera ingestion + anonymous event detection	Creates the physical-world signal.	Facial identity, biometrics
Shelf void detection on 10-20 SKUs	Direct operational value.	All-SKU recognition
Basic POS + inventory correlation	Connects activity to commerce.	Complex ERP marketplace
Action queue	Creates manager workflow.	Consumer-facing app
Experiment ledger	Creates differentiated learning loop.	Full autonomous execution
Matched test/control analytics	Establishes credible ROI.	Enterprise MMM
Store policy memory	Starts compounding advantage.	Large benchmark network

The single best MVP workflow

PILOT LOOP

Select one shelf/category -> detect an operational or conversion problem -> generate 2-3 possible causes -> choose one reversible intervention -> run a test/control window -> measure incremental result -> store the learning -> recommend the next action.

MVP success conditions

- Events are reliable enough for the chosen 10-20 SKUs/zones.
- Manager receives fewer, better alerts rather than a flood of alerts.
- At least one intervention can be measured with a defensible before/after or test/control design.
- The store team can execute the test without a heavy hardware change.
- The experiment ledger shows a credible link from signal -> action -> outcome.

08 | TEN DIFFERENTIATED USE CASES

The new system learns what works - not just what looks abnormal

Start with measurable, reversible interventions.

Use case	Trigger	Test	Measured result
Shelf position test	“A high-interest SKU is underperforming despite availability.”	Move one bay vs control.	Units, margin, basket attachment.
Bundle adjacency test	“Customers buy A and B, but rarely see them together.”	Place together in one test zone.	Attachment rate, units/basket.
Restock timing test	“The same shelf empties repeatedly at 6-8 PM.”	Replenish earlier vs current routine.	Stockout minutes, sales protected, labor.
Endcap message test	“Traffic is high but conversion is weak.”	Change message/display in test zone.	Dwell, pick, conversion.
Queue staffing test	“Queue pain appears before sales peak.”	Shift one staffing block.	Wait time, throughput, labor cost.
Assortment test	“A long-tail item gets attention but low purchase.”	Swap one SKU in a controlled zone.	Category margin, substitution, waste.
Fresh markdown timing	“Expiry risk and traffic do not match.”	Test earlier/later markdown window.	Waste, sell-through, margin.
Cold-zone activation	“Zone traffic is lower than expected.”	Change category placement/wayfinding.	Traffic, dwell, conversion.
Promotion proof	“Promo seems to lift sales.”	Create comparable control period/zone.	Incremental units and margin.
Associate task optimization	“Many tasks are low value at peak.”	Reprioritize task queue.	Minutes saved, availability, execution.

DESIGN PRINCIPLE

A use case becomes strategic only when the system can estimate what would happen without the intervention. Correlation creates a lead; controlled evidence creates a learning asset.

19 | ADDITIONAL “FUTURE OF RETAIL” IDEAS

Optional bets around the core learning engine

These are expansion hypotheses - not MVP requirements.

Idea	Concept	Potential value
Store Memory Graph	A searchable causal memory of what worked, failed and why at each store.	Faster future decisions and lower repeat mistakes.
Synthetic Shopper Lab	AI shopper agents estimate likely path/choice changes before physical deployment.	Cheaper pre-screening of layout ideas.
Constraint-Aware Actions	System knows labor, safety, space, promo and inventory constraints before suggesting a test.	More executable recommendations.
Experiment Marketplace	Reusable templates for common retail tests, with anonymized benchmark priors.	Faster deployment across stores.
Outcome-Based Billing	Optional fee for verified value created.	Aligns vendor and retailer incentives when attribution is strong.
Supplier Learning Network	Permissioned cross-store category evidence for suppliers.	New B2B revenue layer.
Shelf-to-Backroom Agent	Links shelf state to backroom inventory and proposes task sequence.	Reduces stockout response time.
Store Autopilot (later)	Under policy guardrails, system can automatically create tasks and schedule tests.	Higher operating leverage.

20 | NOVELTY SCORECARD

Where the company should and should not claim uniqueness

A disciplined differentiation statement is stronger than a “nobody has this” claim.

Capability	Already publicly established?	RetailVision 2.0 stance
Traffic / heatmaps	Yes - RetailNext and others. [W2]	Table stakes.
Pick / put-back / shopper journeys	Yes - Trigo. [W1]	Table stakes / integration input.
Shelf out-of-stock detection	Yes - Vusion and others. [W3]	Table stakes / input.
POS + inventory fusion	Yes - multiple platforms. [W2-W5]	Required foundation.
Store digital twin / simulation	Yes - research and public retail examples. [W6-W7]	Useful pre-screen, not unique alone.
Experimentation / incrementality	Yes - specialist platforms exist. [W9]	Required analytical capability.
Automated closed-loop: detect -> choose experiment -> verify -> update store policy	Not clearly verified as a single retrofit retail product in this review.	Primary white-space hypothesis; validate aggressively.

DEFENSIBLE WORDING FOR A PITCH

“We are not inventing computer vision for retail. We are packaging existing sensing into a closed-loop learning system that measures which physical-store actions cause incremental value and makes the next decision better.”

24 | FINAL RECOMMENDATION

Build the learning loop

The differentiated company is one that gets smarter from every real store decision.

The original RetailVision concept is commercially sensible, but the public market already contains substantial overlap with its basic feature set. The best strategic update is therefore not “add more computer vision.” It is to make the product responsible for a different outcome: verified, store-specific learning about what actions cause improvement.

That means the heart of RetailVision 2.0 is not the camera, dashboard, digital twin or LLM. The heart is the intervention record and the learning loop: what the system observed, what it believed, what it changed, what happened, and what the store now knows.

THE SINGLE SENTENCE TO REMEMBER

SEE -> DIAGNOSE -> TEST -> VERIFY -> LEARN -> ACT BETTER

Bottom line for the manager

Do not fund a giant “AI supermarket platform” from the idea alone. Fund a tightly controlled pilot that tests whether a closed-loop store learning engine can create measurable value with existing infrastructure. If it works, the data asset and policy memory give the startup a clearer path to defensibility than a generic retail-vision dashboard.

SOURCES & EVIDENCE

External research checked for this updated blueprint

Research date: September 17, 2026

The sources below are used to distinguish already-established market capabilities from the proposed white-space hypothesis. They are not endorsements of RetailVision AI, and company descriptions are based on their own public materials unless otherwise noted.

Source	What it supports	Reference
[W1] Trigo Retail - Retail Intelligence	Traffic patterns, shopper journey, product interaction, pick/put-back, decision sequence, placement impact, anonymous tracking.	https://www.trigoretail.com/retail-intelligence/
[W2] RetailNext - Traffic Analytics / Insights	Traffic, conversion, full shopper path, AI prediction, prescriptive recommendations and test-and-learn.	https://retailnext.net/platform/traffic-analytics ; https://retailnext.net/platform/insights
[W3] Vusion - Computer Vision & Store Operational Excellence	Shelf gaps, planograms, inventory, prioritized tasks, connected-store execution.	https://www.vusion.com/technologies/computer-vision-ai ; https://www.vusion.com/solutions/store-operational-excellence
[W4] Shopic - Platform / Store Analytics	Shopper decisions, live inventory, promotions, shelf intelligence and store analytics.	https://www.shopic.co/platform/
[W5] Vori - Grocery OS	End-to-end grocery operating system positioning including payments, shelf tracking, ordering, pricing and promotions.	https://www.vori.com/news/vori-raises-22m-series-b
[W6] Journal of Contemporary Marketing Science (2026) - AI-enabled store digital twins	Conceptual store digital twins combining physical/digital data and human-in-the-loop optimization.	https://www.sciencedirect.com/org/science/article/pii/S2516748025000119
[W7] Lowe's Innovation Labs - Store Digital Twin	Public example of store visualization and simulated placement/layout analysis.	https://www.lowesinnovationlabs.com/projects/store-digital-twin
[W8] McKinsey (2026) - Agentic AI in retail merchandising	Agentic AI can automate merchandising analysis, generate scenarios and learn from cycles; stresses governance and phased pilots.	https://www.mckinsey.com/industries/retail/our-insights/merchants-unleashed-how-agentic-ai-transforms-retail-merchandising
[W9] Circana - Liquid Testing	In-store incrementality testing and causal measurement at store level.	https://www.circana.com/liquid-testing
[Source blueprint] User-provided RetailVision AI Manager Review Draft	Original concept, MVP, pricing hypotheses, ROI framing, architecture and pilot plan.	Provided in the conversation as RetailVision_AI.pdf

IMPORTANT LIMITATION

No public-web review can prove worldwide non-existence. The novelty language in this document is intentionally framed as a "white-space hypothesis" and "not clearly publicly documented in this review," not as a guarantee that no competitor, startup, internal team or patent has already developed similar technology.